9th

Monday, September 14, 2026 9:56 AM

Error Analysis

a priori ~~Analysis~~ error:

① Energy norm

② L^2 - norm

A posteriori error

① ~~e~~ energy norm

$$\|e\|_E^2 = \int_0^1 a e' e' dx$$

The from FEM

$$\|e\|_E^2 = \int_0^1 f(e - \pi_h e)\, dx - \int_0^1 a v'(e - \pi_h e)' dx$$

but from interpolation

$$\| h^{-1}(e - \pi_h e) \|_a \leq C \|e'\|_a \, O(h)$$

$O(h^r)$

Note: roughly speaking $\beta(U) = O(1)$



In practice, can do this as follows

(1) on some initial coarse mesh compute the FEM solution

(2) calculate on each element the residual.

$$f_j^r = (x_j - x_{j-1})^r \int_{x_{j-1}}^{x_j} \frac{1}{a(x)} R^r(U)\, dx$$

where, $R(U) = f + (aU')'$

want, $E^2 = \sum_{j=1}^{M+1} E_j^2 \leq \frac{\epsilon^2}{c^2}$

If $E_j^2 \leq \frac{\epsilon^2}{c^2(M+1)}$

(3) or calculate the solution on new mesh.

then repeat → (2) + (3)